Problem 1: Given two words *word1*, and *word2*, find the minimum number of operations required to convert *word1*, to *word2*. You have the following 3 operations permitted on a word:
1. Insert a character 2. Delete a character 3. Replace a character

Problem 2: Consider the problem of what strategy to take when moving your fingers over a keypad to dial a telephone number. You want to dial using two fingers, which begin respectively on the $*$ and # keys. The objective is to devise a strategy so that the fingers take the least amount of effort overall for dialing a full n digit number. You may assume that the effort involved in moving one finger from one key to another is proportional to the distance between the two keys, and that a finger will remain hovering over a key until it is moved to the next key. (Use the distance between keys as the Euclidean distance, and round to the nearest 1/2 to make the arithmetic easy. Eg. take the distance between keys next to each other as 1 unit if it is a vertically or horizontally next to, and 1.5 units if diagonally i.e. $\sqrt{2}$ rounded to the nearest 1/2.)

*[**Hint:** begin with the entries corresponding to no digits remaining to be dialed. Next consider how to calculate the total effort starting at some finger configuration and, say n digits left to dial; how would you calculate that if you already know the least amount of effort for any finger configuration and $n - 1$ digits left to dial?]*